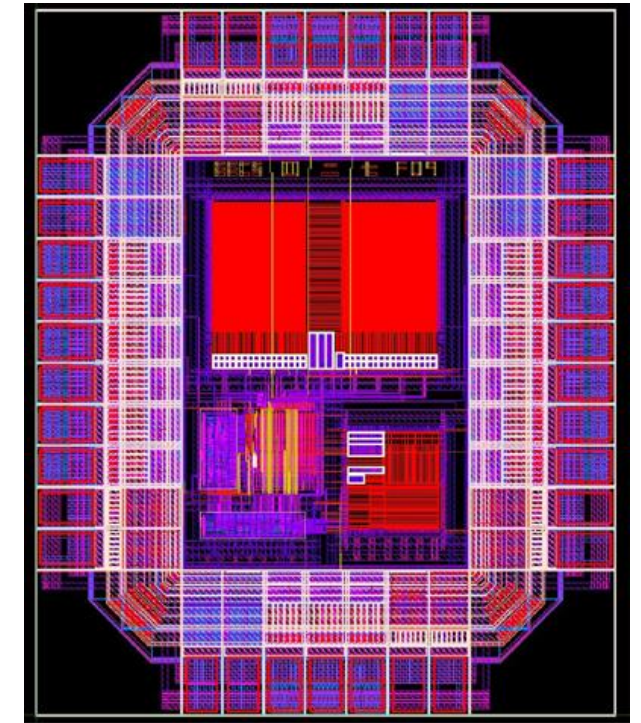
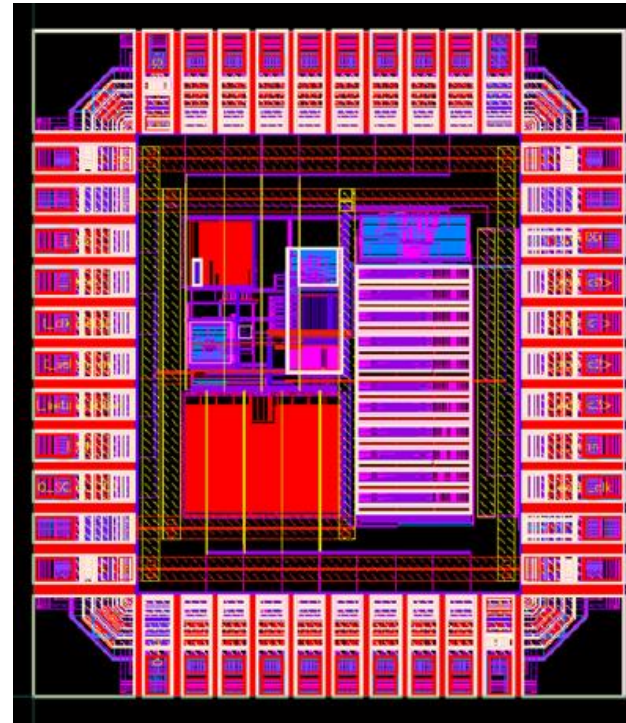
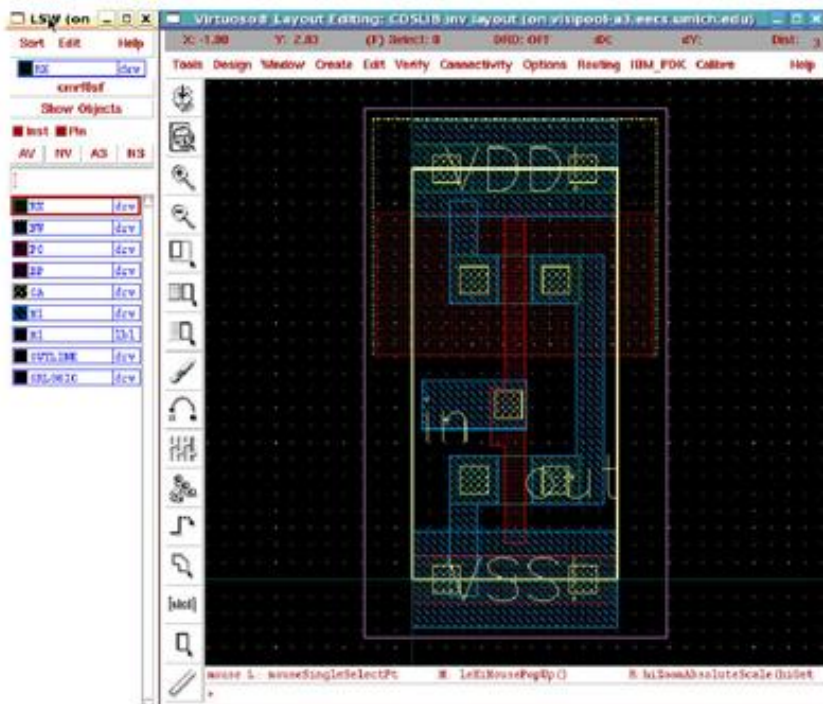


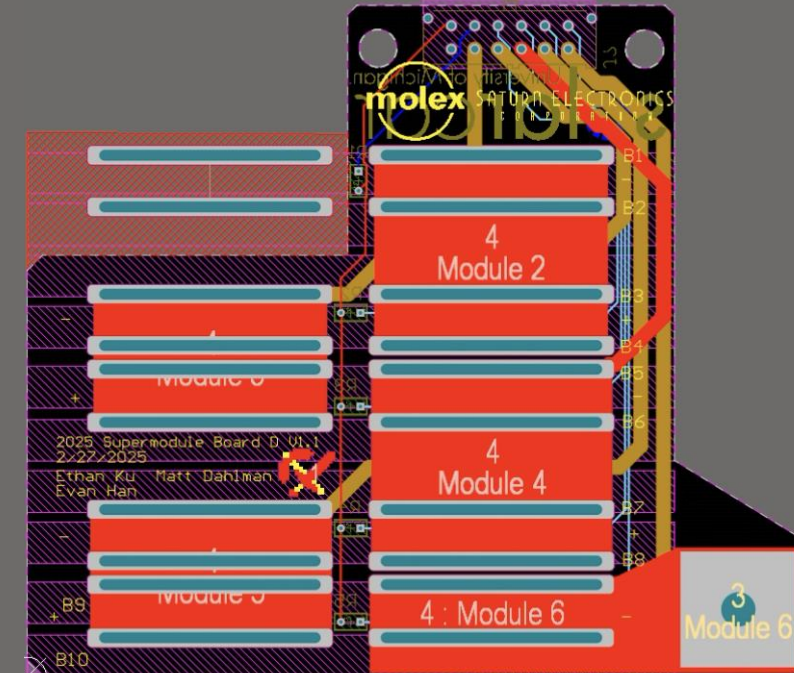
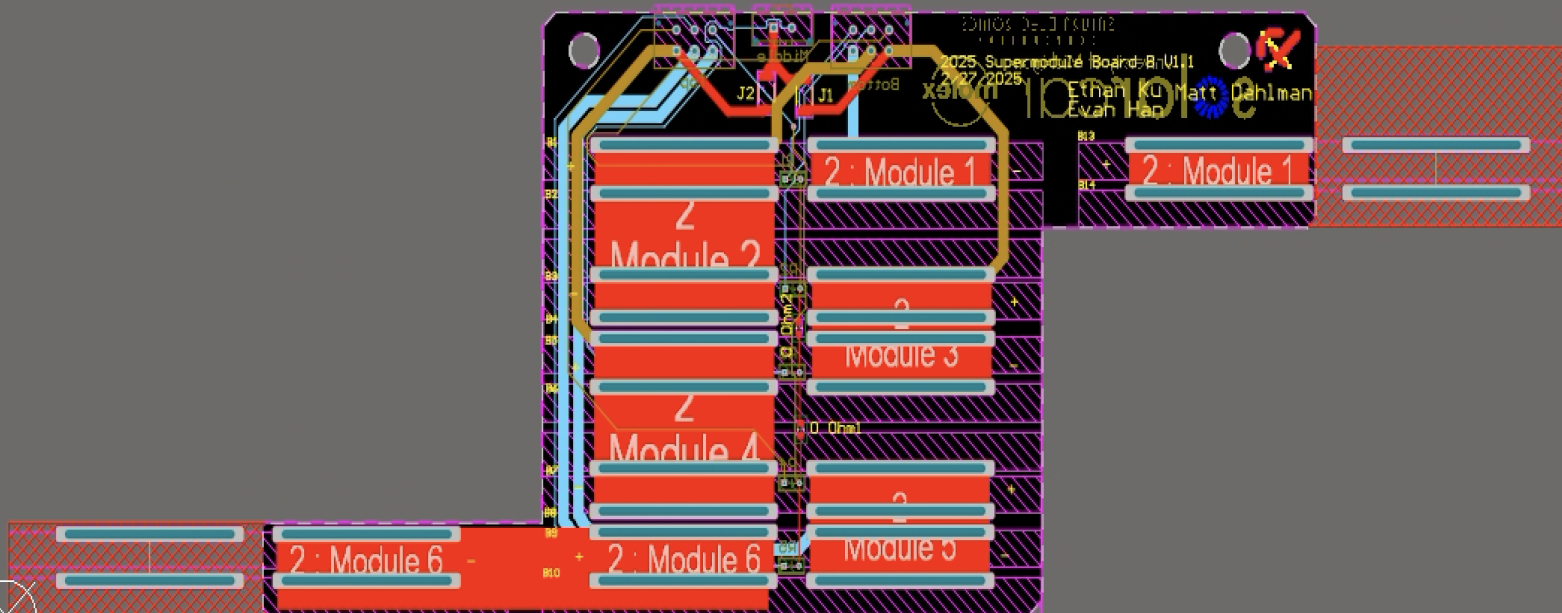
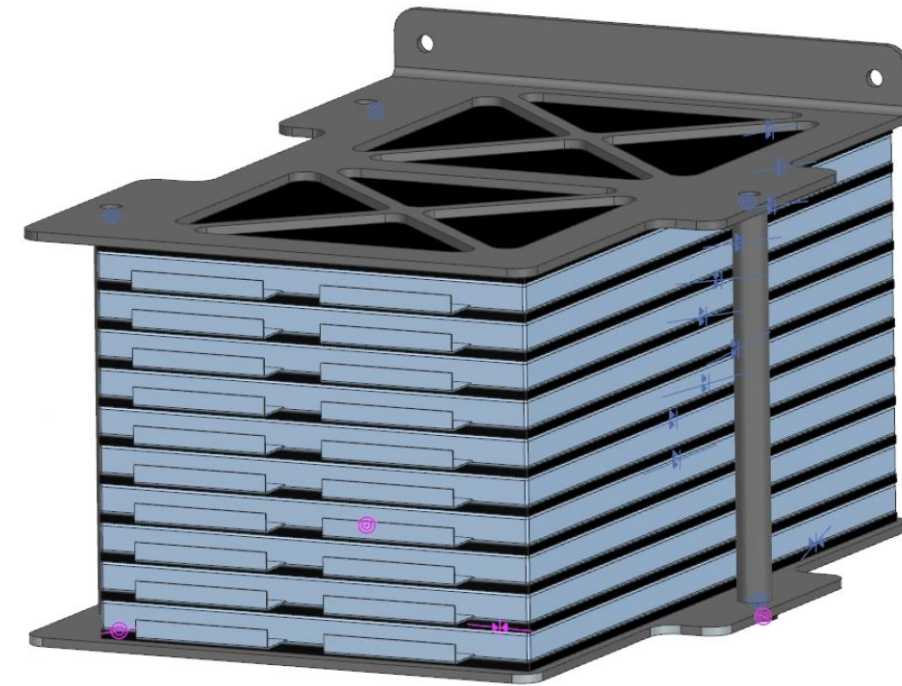
RISC-V Microprocessor

- Created 16-bit, 65nm RISC-architecture microprocessor, achieving a 500 Mhz clock frequency with 53, 600.88 μm^2
- Virtuoso for transistor schematic and layout, Maestro for speedpath, NCVerilog for verification, and DRC and LVS
- Integrated on-chip SRAM caching with CPU to reduce memory-access latency between CPU and external DRAM

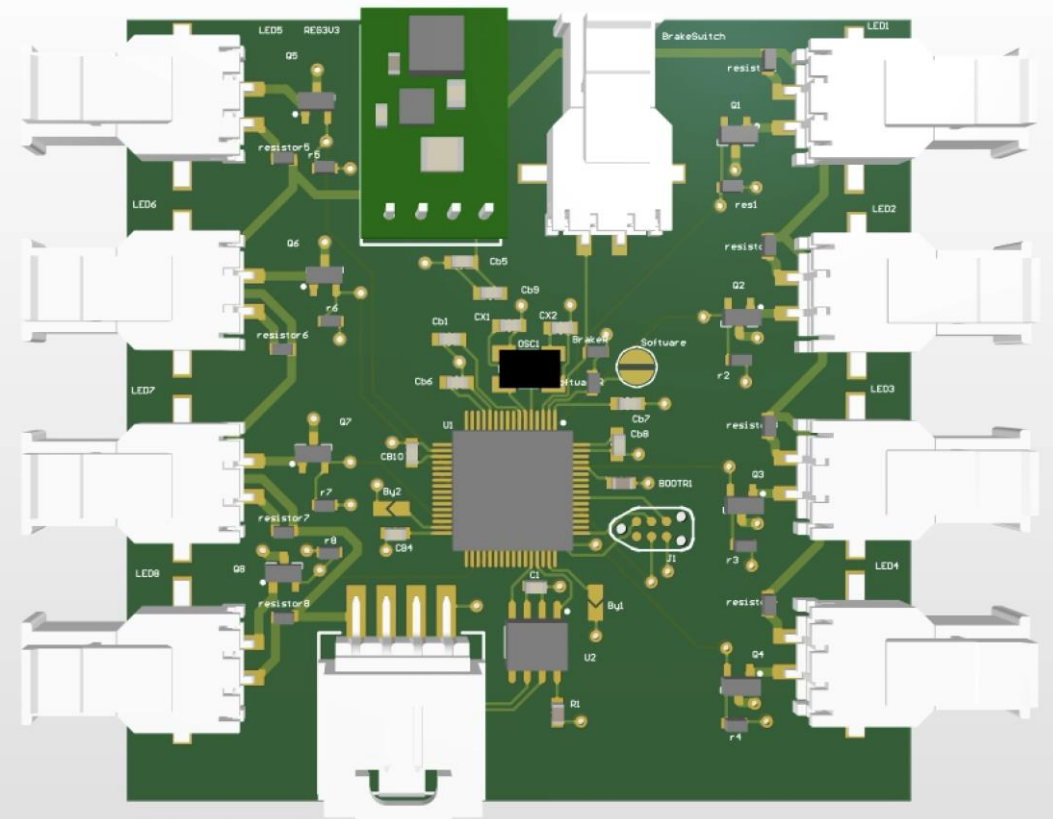
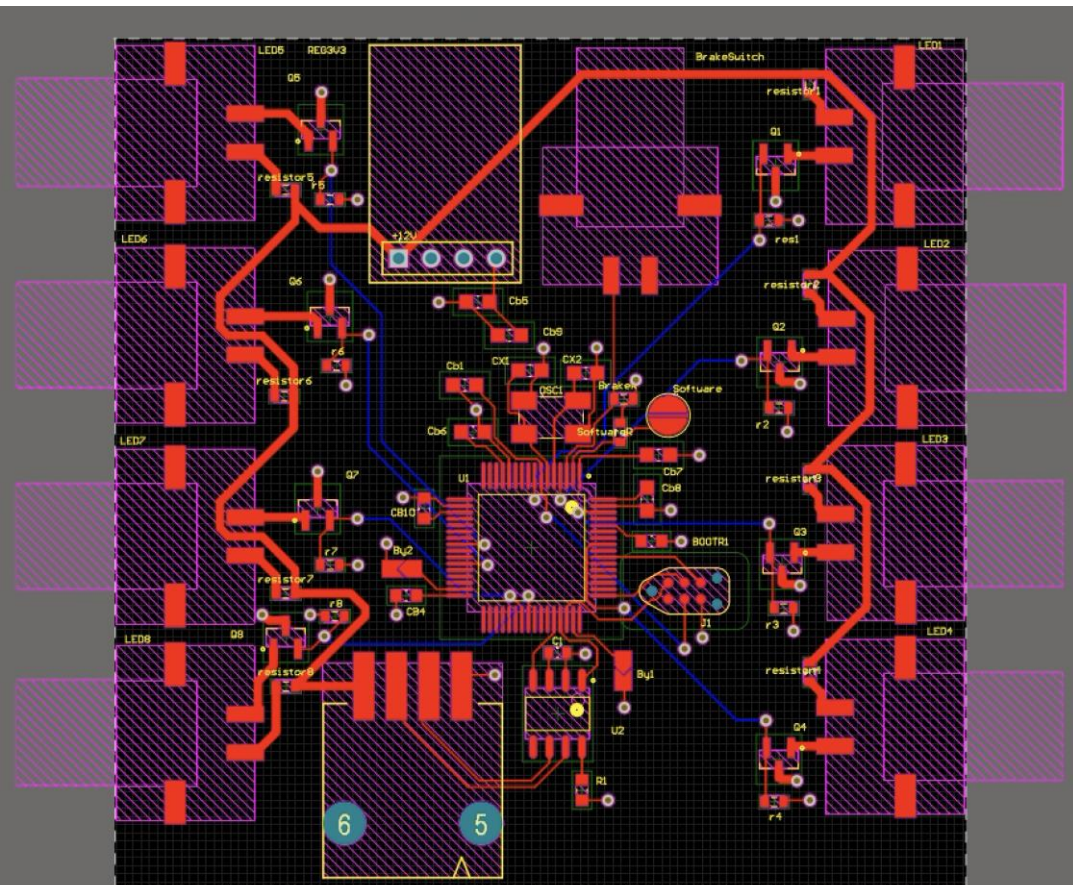
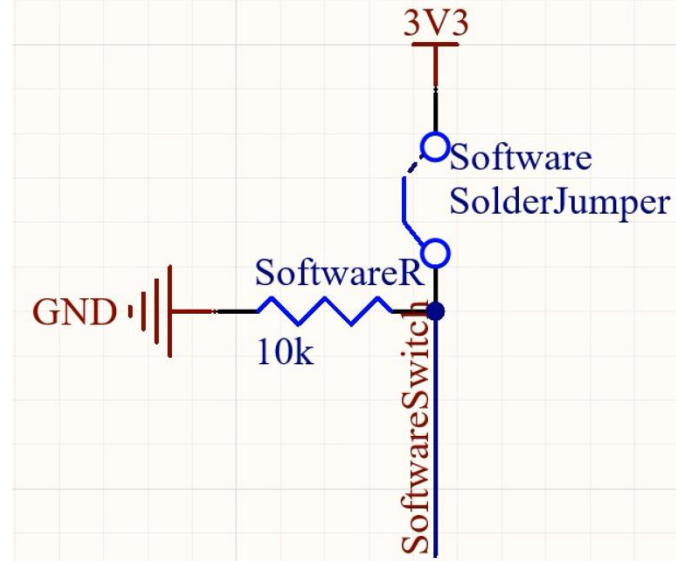


Note: Official images are under NDA, these are publicly accessible images from the UMich website which show the same RISC-V Microprocessor we create

- Battery Supermodule design with geometric improvement for improved safety by eliminating high voltage copper bus bars
- 100% functionality of car battery: series-parallel connections create capacity for travelling 3000+ km in Bridgestone World Solar Challenge (7th place)
- Teamcenter/NX CAD dimension fitting



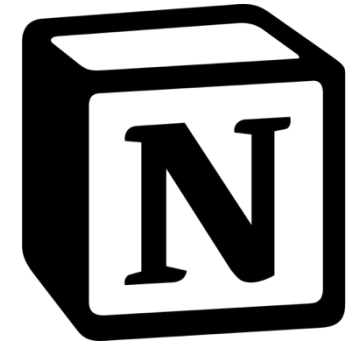
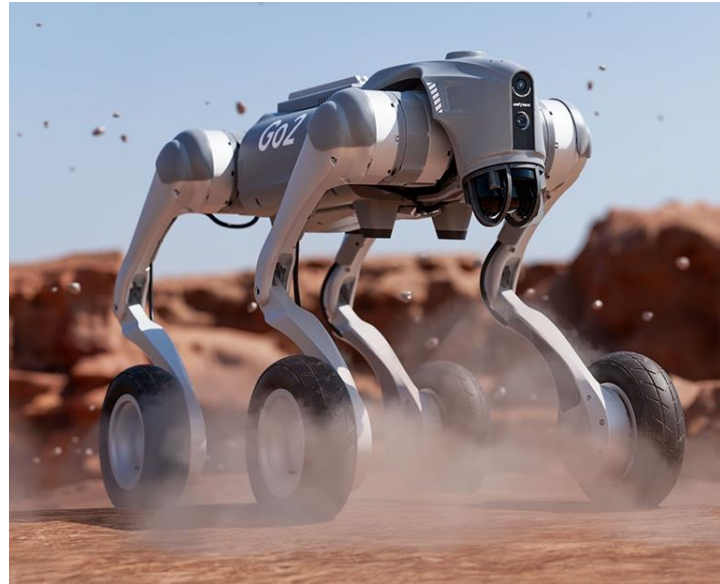
- Brake Light board with solder jumper design to set software logic for front or back brakes, pulling to GND or connecting based on solder connection
- 50% decreased inventory clutter improving limited travel team storage space





Research Assistant

- Led design of PDB on mobile robot chassis by reviewing and implementing past practices and new component requirements, ensuring power needs for 4 new components, and verifying criteria for continued V1 components
- Integrating 2 new sensors on robot chassis for real-time localized navigation via SLAM, communicating with ROS2
- Implemented Notion with team of 5-10 by reviewing old Google Drive inefficiencies and matching identified team needs with modern management software for 85% increased information access and task deployment efficiency

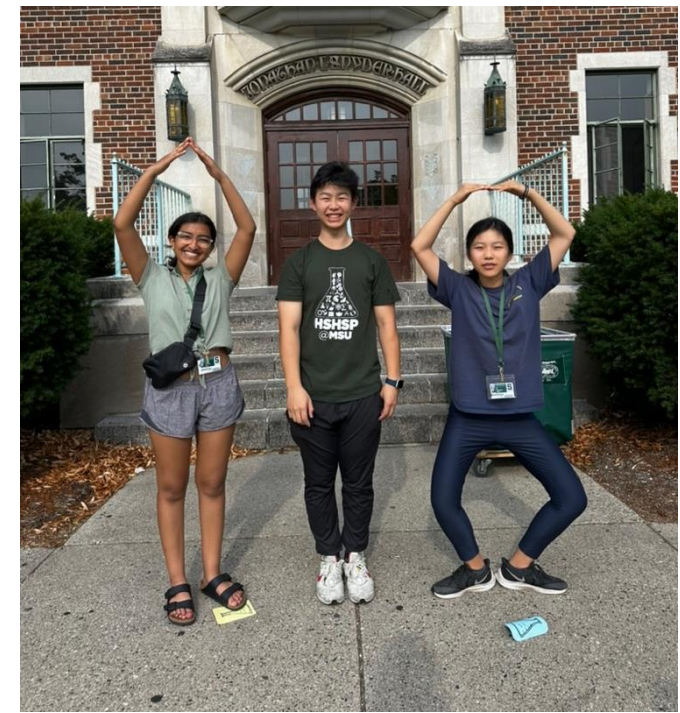




HSHSP

Research Assistant

- HSHSP Program (2023)
- Publication: D. T. Islam, E. Telli, N. Telli, H. Fatteh, I. Ma, E. Ku, S. Kotaru, W. Hanim, P. Hatzinger, B. Upham, M. Williams, G. Zylstra, D. Fennell, A. Cupples, S. Hashsham. (2023). “Targeted Sequencing Panel to Characterize the Respiration of Polychlorinated Dibenzo-p-dioxins and Dibenzofurans by Dehalococcoides mccartyi Strains”. Annual Superfund Research Project Meeting, Albuquerque, New Mexico, 2023



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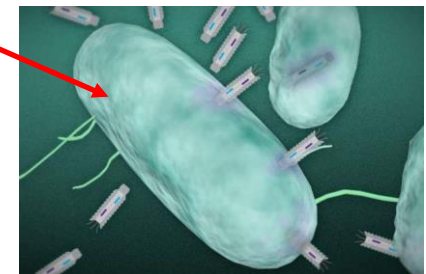
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Team Member (Class)

- Developed 2 full-scale remote controlled hovercraft with team of 5, capable of delivering payloads up to 100g
- Led integration of MOSFET connections to battery, directional servo firmware, and LEDs per nautical regulations

